

MD3B v.1.0 – User guide

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1. Introduction

In the domain of neuropharmacological drug discovery, predicting the capacity of a small molecule to permeate the blood-brain barrier is a critical milestone. Molecular dynamics simulations, coupled with enhanced sampling techniques such as umbrella sampling, provide a rigorous theoretical framework for quantifying passive membrane transport. The primary objective of such simulations is to extract position-dependent thermodynamic and kinetic parameters to compute the macroscopic effective permeability coefficient.

The software package under review consists of a Python implementation designed to parse trajectory data from GROMACS molecular dynamics simulations, compute position-dependent diffusion coefficients, apply free energy corrections for ionizable compounds, and output empirical pharmacokinetic metrics such as LogBB. This comprehensive report provides an exhaustive evaluation of the consistency between the executing Python codebase and its accompanying technical documentation. Through a detailed biophysical and algorithmic audit, the analysis contrasts the theoretical claims against the executed programming logic. The findings establish that the core physicochemical equations, execution models, file selection paradigms, and adaptive statistical algorithms are accurately represented and highly consistent between the documentation and the codebase. However, a divergence still exists regarding the final classification outputs.

The application takes GROMACS umbrella sampling trajectory files and Potential of Mean Force (PMF) data to compute:

- Local diffusion coefficient profiles D(z)
- Effective permeability (Peff)
- LogBB values (logarithm of brain-to-blood concentration ratio)

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2. Theoretical Foundations

To understand the algorithmic design of the software, it is necessary to examine the underlying biophysical theory it attempts to execute. The software computes permeability based on the Inhomogeneous Solubility-Diffusion Model (ISDM), originally formalized for lipid bilayer transport by Marrink and Berendsen (*The Journal of Physical Chemistry,* 98, 1994, 4155–4168). Classical homogeneous models assumed a uniform hydrophobic core. However, the ISDM recognizes that a lipid membrane exhibits profound spatial heterogeneity. A permeating solute transitions through distinct dielectric environments: the bulk aqueous phase, the highly polar and ordered lipid headgroup region, the dense upper acyl chains and the disordered hydrophobic core at the bilayer center.

Under the ISDM framework, the steady-state flux of a molecule across this anisotropic medium is governed by two spatially dependent variables defined along the membrane normal (the z-axis): the potential of mean force (PMF), represented as *G(z)* and the local position-dependent diffusion coefficient, represented as *D(z)*. The effective permeability coefficient is calculated as the inverse of the integrated resistivity across the entire thickness of the membrane. The resistance at any given plane is exponentially weighted by the free energy barrier and inversely proportional to the local diffusivity.

Note on permeability theory: The effective permeability can be calculated in the inhomogeneous solubility diffusion model context by the D_Z from the Hummer method and $\Delta G_{(Z)}$ from PMF by the Smoluchowski equation:

$$P_{eff} = \left[\int_{z_{blood}}^{z_{brain}} \frac{\exp(\Delta G(z)/k_B T)}{D(z)} \, dz \right]^{-1}$$

where:

$\Delta G_{(Z)}$**:** The free energy profile across the membrane.

D_Z**:** The position-specific diffusion coefficient.

$k_B T$: Thermal energy.

2.1. The Physics of Permeation: The Inhomogeneous Solubility-Diffusion Model (ISDM)

To fully capture the anisotropic mechanics of the lipid bilayer, researchers rely on explicit, atomistic Molecular Dynamics (MD) simulations. Within MD, the solute, the specific lipid molecules, and the surrounding water are all represented explicitly, allowing for the precise computation of position-dependent forces. However, translating raw atomic coordinates into a macroscopic permeability coefficient requires a rigorous statistical mechanical framework, known as the inhomogeneous solubility-diffusion model (ISDM).

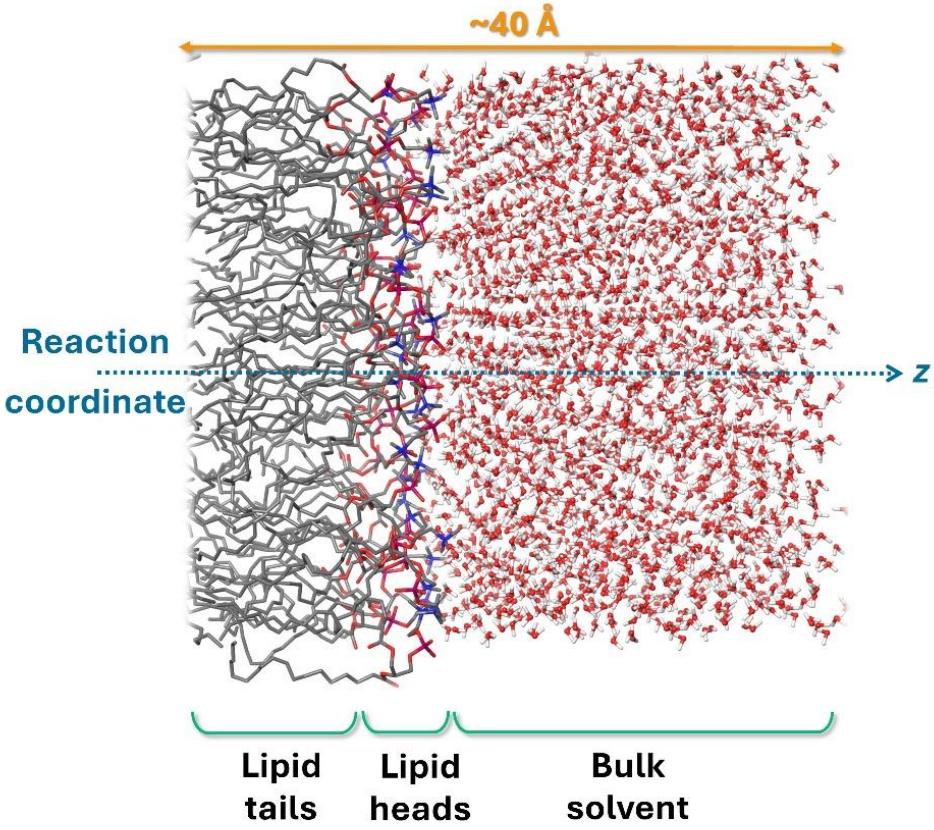


Figure 1. The components of a typical MD system involved in the ISDM approach calculations and successive LogBB prediction.

2.2. Beyond Fick First Law and Homogeneous Slabs

Classical models of diffusion treat the membrane as a homogeneous hydrophobic slab. The first law of Fick defines the flux (*J*) of particles across this slab as proportional to the concentration gradient:

$$J = -D\frac{\Delta C}{2L}$$

where *D* is a constant diffusion coefficient and *2L* is the membrane thickness. When incorporating the partition coefficient (K_p) of the solute into the membrane, the classic homogeneous solubility-diffusion model yields a straightforward permeability equation:

$$P = \frac{K \cdot D}{2L}$$

This relationship forms the basis of the rule of Overton and Meyer, which suggests that membrane permeability correlates directly with lipophilicity (*ChemPhysChem*, 10, 2009, 1405-1414). However, treating the lipid bilayer as a homogeneous slab leads to severe theoretical artifacts. A highly lipophilic, bulky molecule may exhibit a massive partition coefficient ($K >> 1$), heavily favoring insertion into the membrane interior. Under the homogeneous model, this guarantees high

permeability. In reality, such a molecule may partition into the lipid core and become permanently trapped there due to the energetic cost of exiting back into the opposing aqueous leaflet.

2.3. The Derivation of Marrink and Berendsen

To rectify this, Marrink and Berendsen revolutionized the field by defining the membrane as a continuously varying medium along the z-axis (the axis normal to the bilayer surface). Based on the Smoluchowski equation for overdamped Langevin dynamics, the ISDM recognizes that both the thermodynamic free energy and the kinetic diffusion rate are heavily dependent on the solute specific depth within the membrane. The partition coefficient is replaced by a position-dependent Potential of Mean Force (PMF), denoted as $\Delta G(z)$. The PMF represents the reversible work required to move the solute from bulk water to depth z. Correspondingly, the diffusion coefficient is replaced by a local, position-dependent diffusion profile $D(z)$.By treating the continuous membrane as an infinite series of thin slices acting as resistors in parallel with respect to transcellular passage, the local resistance to permeation $R(z)$ at any depth is given by:

$$R(z) = \frac{\exp(\beta \Delta G(z)))}{D(z)}$$

where $\beta = (k_B \cdot T)^{-1}$, k_B is the Boltzmann constant, and T is the absolute temperature. The global resistance of the membrane is simply the integral of the local resistances across the bilayer bounds (from $-L$ to L). Because intrinsic permeability is the inverse of global resistance, the ISDM mathematically defines the effective permeability coefficient (P_{eff}) as:

$$P_{eff} = \left(\int_{-L}^{L} \frac{\exp(\beta \Delta G(z)))}{D(z)} dz \right)^{-1}$$

This elegant formulation proves that overall permeability is mathematically bottlenecked by the highest regions of local resistance. A highly polar molecule will face a massive free energy spike ($\Delta G(z) >> 0$) at the hydrophobic core, driving resistance to infinity. A bulky, hydrophobic molecule might have a negative PMF in the core, but if its size causes it to become entangled in the densely packed lipid acyl chains, its local diffusion $D(z)$ will approach zero, similarly driving resistance to infinity. The ISDM thus captures both the thermodynamic partitioning and the kinetic traversal constraints perfectly.

2.4. The ISDM Formalism

While the ISDM equation is theoretically robust, executing the molecular dynamics required to accurately compute $\Delta G(z)$ and $D(z)$ for complex pharmaceuticals is exceedingly difficult. Spontaneous membrane crossing is a rare event that occurs on microsecond to millisecond timescales, well beyond the reach of standard unbiased MD simulations. The current standard for operationalizing the ISDM in modern drug discovery was paved by Carpenter, Lightstone, and colleagues in their publication, "A Method to Predict Blood-Brain Barrier Permeability of Drug-Like Compounds Using Molecular Dynamics Simulations" (*Biophysical Journal,* 107, 2014, 630–641). This seminal work combined enhanced sampling techniques, specific lipid parameterizations, and advanced statistical analyses into a highly reproducible protocol. The Python framework introduced

in this article is specifically architected to automate every mathematical step of the Carpenter formalism.

2.5. The Membrane Mimetic and Enhanced Sampling Strategy

To standardize predictions, the Carpenter formalism utilizes a simplified, pure lipid bilayer as a surrogate for the BBB. Specifically, 1,2-dioleoyl-*sn*-glycero-3-phosphocholine (DOPC) is employed. A typical simulated system contains approximately 72 DOPC lipid molecules, 5.124 water molecules and a single drug compound, resulting in a system size of roughly 19.300 explicit atoms. The DOPC bilayer is equilibrated for up to 500 ns to ensure proper liquid-disordered state behavior before the drug is introduced. The resulting membrane must display structural properties consistent with experimental measurements, including an area per lipid of approximately 60 $Å^2$ (compared to ~30 $Å^2$ for densely packed sphingomyelin or cholesterol) and a bilayer thickness of roughly 47.5 Å. To overcome the rare-event barrier, the formalism employs steered molecular dynamics (SMD) combined with umbrella sampling. The drug molecule is artificially pulled through the DOPC bilayer along the *z*-axis using a harmonic spring. From this pulling trajectory, coordinates are extracted at regular spacing intervals (e.g., every 0.1 nm). Numerous independent MD simulations (the umbrellas) are then executed. In each window, the drug center of mass is harmonically restrained to its specific *z*-coordinate but is allowed to freely rotate and sample lateral conformational space over nanosecond timescales.

2.6. Extracting the Potential of Mean Force via WHAM

The biased positions recorded during the umbrella sampling windows result in a series of overlapping Gaussian probability distributions. The unbiased free energy profile $\Delta G(z)$ is extracted from these distributions using the weighted histogram analysis method (WHAM). The resulting PMF profile is typically symmetric around the bilayer center ($z = 0$). The profile is normalized to zero in bulk water (e.g., at $z = \pm 3$ nm). Carpenter et al. demonstrated that integrating the resistance across the entire normalized PMF landscape, rather than relying on a single free energy delta at the bilayer center, protects the permeability calculation from being overly skewed by localized sampling errors. Furthermore, the method was proven insensitive to minor variations in the integration boundary.

2.7. Deriving Diffusion from Bayesian Position Autocovariance

Calculating the position-dependent diffusion coefficient $D(z)$ for an artificially restrained molecule is one of the most mathematically intense components of the formalism. Standard diffusion calculations relying on mean squared displacement (MSD) are invalid because the harmonic restraint inherently limits the particle displacement over time. To solve this, the Carpenter formalism leverages a theoretical framework developed by Hummer, which extracts diffusion rates from the thermal position fluctuations within the harmonic restraint (*New Journal of Physics,* 7, 2005, 34). Hummer demonstrated that the diffusion coefficient is related to the position autocovariance function of the *z*-coordinate by the relationship:

$$D(z) = \frac{\mathrm{var}(z)^2}{\int_0^\infty C\,(\tau) d\tau}$$

where *var(z)* is the variance of the simulated *z*-position (which is equal to the autocovariance at lag time zero), and *C(τ)* is the autocovariance function at lag time *τ*. In practice, the autocovariance curve of the restrained particle decays approximately exponentially over time. The denominator can therefore be solved by numerically fitting the natural log of the autocovariance data to extract a characteristic decay time. Because this mathematical extraction must be performed independently across up to a hundred separate umbrella windows to generate a continuous *D(z)* profile, it constitutes a massive computational bottleneck when attempted manually.

2.8. Physiological Protonation Corrections (The MacCallum Penalty)

A profound limitation of many early MD studies was the failure to account for multiple protonation states. At the physiological blood pH of 7.4, many drug molecules exist in an equilibrium between ionized and neutral species. The charged species is highly solvated in the aqueous plasma but faces a massive, often insurmountable dielectric penalty upon entering the lipid core. Consequently, transcellular permeation is almost entirely driven by the neutral species, even if it comprises less than 1% of the total drug population in the blood. The Carpenter formalism elegantly resolves this by mandate: the complete umbrella sampling, WHAM and Hummer diffusion protocols must be executed twice, once for the charged and once for the neutral species. This yields the parallel resistance profiles $R_c(z)$ and $R_n(z)$, respectively. Crucially, the PMF of the neutral species must be adjusted to account for the free energy required to deprotonate (or protonate) the dominant charged species in the bulk water before it can enter the membrane. Utilizing the method of MacCallum, Bennett and Tieleman (*Biophysical Journal,* 94, 2008, 3393–3404; *The Journal of General Physiology,* 129, 2007, 371–377), this neutralization penalty is derived from the experimental or calculated pK_a using the Henderson-Hasselbalch equation:

$$\Delta G_{\text{neutralization}} = 2.303 \cdot R \cdot T \cdot |\text{pH} - \text{p}K_a|$$

where R is the ideal gas constant and T is temperature. This constant offset is added to the entire $\Delta G_n(z)$profile. The total effective permeability is then calculated by summing the flux contributions of both pathways:

$$P_{eff} = \left(\int_{-L}^{L} R_c\,(z)dz\right)^{-1} + \left(\int_{-L}^{L} R_n\,(z)dz\right)^{-1}$$

By combining the pH adjustment with the complete ISDM integral, the methodology can generate remarkable quantitative results, toward establishing the ISDM MD approach as a highly predictive, physics-based alternative to the extremely demanding experimental screening.

The source code implements this theoretical model with high fidelity. The software extracts the PMF array, interpolates it to match the spatial grid of the calculated diffusion coefficients using cubic spline interpolation and computes the local resistivity array. The final integration step is performed using the simpson function from the scipy.integrate module. It is notable that the scipy.integrate.simpson rather than the older simps function was utilized for adherence to modern library standards, as the simps function was deprecated in recent releases and slated for complete removal in SciPy version 1.14.0. This architectural decision ensures the long-term computational stability of the integration engine.

A note on the neutral-PMF correction procedure used for ionizable compounds

- Compute charged-species resistivity $R_c(z) = exp\,(G_c\,(z)/k_B\,T)/D_c\,(z)$ from the ionized PMF and its umbrella-derived $D_c(z)$.
- Compute the pKa-dependent neutralization shift $dG = (pH - pKa)k_BTln(10)$ (for acids) or $dG = (pK_a - pH)k_BTln(10)$ (for bases), then offset the neutral PMF: $G_n^{corr}(z) = G_n(z) + dG$.
- Estimate the neutral diffusion profile $D_n(z)$ from neutral umbrella trajectories; if missing, fall back to $D_c(z)$.
- Compute neutral resistivity $R_n(z) = exp\,(G_n^{corr}\,(z)/k_B\,T)/D_n\,(z)$.
- Combine resistivities for two-state permeation: $R_{\text{comb}}(z) = \frac{R_c(z)R_n(z)}{R_c(z)+R_n(z)}$.
- Integrate to get corrected permeability: $P_{\text{eff}} = 1/\int R_{\text{comb}}(z)dz$, then calculate corrected LogBB from P_{eff}.

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3. Implementation

The software integrates a sophisticated graphical user interface with an advanced level of desktop application engineering constructed utilizing the standard Python Tkinter library. To mitigate common visual artifacts associated with dynamic plotting and window resizing, custom native Windows API calls were implemented via the ctypes module. Specifically, functions such as _user32.SetWindowLongPtrW and _dwmapi.DwmSetWindowAttribute are utilized to configure native fast background erasure and prevent unmap cascades during window minimization.

3.1. Threaded Concurrency Model

The application relies on the threading module to manage intense molecular dynamics data processing without blocking the GUI event loop. When the simulation is initiated, the software instantiates a single daemon thread targeting the computational worker function. Because Python is constrained by the GIL, multiple threads cannot execute native Python bytecodes simultaneously. However, the performance penalty is largely mitigated in this specific application because the core mathematical operations, such as Fast Fourier Transforms and linear regressions, are delegated to C extensions within the NumPy and SciPy libraries, which routinely release the GIL during execution. To safely communicate between the background calculation thread and the main Tkinter thread, a thread-safe queue.Queue() mechanism is implemented. The background thread places log messages and graphical update lambda functions into memory queues, which are systematically flushed by the main thread using the root.after() scheduling method.

3.2. Automated Directory Traversal

The workflow for locating and importing necessary GROMACS trajectory (pullx.xvg) and potential of mean force (pmf.xvg) files is carefully aligned between the code and the theoretical foundation. **The GUI provides a single browse button for defining the root directory where each drug simulation files should be placed.** Upon selection, the software automatically executes a directory traversal algorithm, relying on the os.listdir and glob modules to search for rigidly defined subdirectories named exactly 'ionized/' and 'neutral/'. The code automatically identifies files ending in *.xvg* that contain the substring "pmf" or "pullx", mapping them to the appropriate internal variables. This automated file discovery system represents a substantial enhancement to the user experience, eliminating manual file selection and drastically reducing the probability of human error, such as inadvertently pairing a neutral trajectory file with a charged free energy profile.

3.3. Estimation of Position-Dependent Diffusion Coefficients

Within the ISDM, calculating the position-dependent diffusion coefficient is arguably the most mathematically complex operation. The software derives local diffusivity utilizing umbrella sampling trajectory data. The theoretical basis of this methodology is attributed to Hummer (*New Journal of Physics,* 7, 2005, 34), where diffusion is computed from the position variance of the harmonically restrained molecule and the integral of its position autocorrelation function (PACF). The fundamental relation posits that the local diffusion coefficient is equal to the variance of the displacement divided by the characteristic autocorrelation decay time. The software must extract this characteristic correlation time from noisy, finite molecular dynamics trajectories.

3.4. Algorithmic Implementation

The active computational pathway relies exclusively on the calculate_tau_least_squares function. This active function employs a highly sophisticated spectral approach. It calculates the autocovariance using a Fast Fourier Transform algorithm (acovf with fft=True from the statsmodels library), establishes a noise floor based on the standard deviation of the terminal lags and isolates the initial exponential decay phase. The function then applies a logarithmic transformation and utilizes scipy.stats.linregress to extract the correlation time from the negative inverse of the resultant slope. This logarithmic linear regression strategy is statistically superior to direct numerical integration, which is highly vulnerable to long-time tail noise and finite sampling errors inherent in molecular simulations. By extracting the decay constant strictly from the high-signal-to-noise initial regime, the software achieves a highly stable estimate of local diffusivity.

3.5. Adaptive Statistical Sampling and Interval Optimization

In the analysis of molecular dynamics trajectories, block averaging is a standard technique used to estimate the standard error of correlated time-series data. The trajectory must be subdivided into discrete intervals or blocks. Determining the optimal size of this interval presents a profound statistical challenge. If the interval is too small, adjacent blocks remain temporally correlated, violating the assumption of independence and leading to a severe underestimation of statistical

error. Conversely, if the interval is too large, the total number of blocks drops precipitously, inflating the variance of the estimate and reducing statistical confidence.

3.6. Heuristic Candidate Scoring Engine

The application features a highly sophisticated, data-driven optimization routine governing interval selection. It accurately establishes the boundary parameters: a minimum floor of 100 points to prevent unstable logarithmic regressions, a maximum cap of 15.000 points to prioritize variance reduction and an exploratory sampling of up to 50 trajectories to map the membrane spatial heterogeneity. Furthermore, it utilizes a multi-objective heuristic candidate scoring engine utilized by the codebase. The algorithm generates an extensive set of candidate intervals, ranging from fixed small increments to dynamic multipliers of the 50th, 75th, and 90th percentile correlation times derived from the sampled trajectories. The engine then evaluates every candidate against the trajectory subsample, calculating a composite score based on a precise mathematical formula. The scoring function weights the percentage of blocks yielding a mathematically positive correlation time at 30%, and the fraction of blocks where the logarithmic regression yields an $R^2 \geq 0.8$ at 15%. Furthermore, it applies a logarithmic bonus for generating larger quantities of blocks, saturating near 250 blocks. Conversely, it applies severe linear penalties if the candidate interval results in fewer than 20 blocks and applies a massive disqualification penalty if the valid fraction drops below 50%. The interval that achieves the highest composite score is selected as the optimal block size for the entire simulation. This implementation is a sophisticated, data-driven optimization routine that balances the physical requirement for temporal decorrelation against the statistical requirement for maximal sample size.

3.7. Selection of interval size for diffusion and autocorrelation time

The Hummer method utilizes the variance and autocovariance of molecular positions from position-restrained MD simulations to calculate local diffusion coefficients. It assumes that the motion of the compound follows an overdamped Langevin process. The variance (V_Z) measures the extent of the drug thermal fluctuations within the restraint potential. The autocovariance function describes how long the molecule "remembers" its previous position. In this model the curve decays exponentially while τ_Z is the characteristic decay time (correlation time). Once the variance and are known, the position-specific diffusion coefficient is calculated using the Einstein-Smoluchowski-like relation.

The interval size variable controls the block length for estimating τ_Z and D. Changing it, affects bias versus variance:

- **Too small:** blocks are shorter than the autocorrelation time, τ_Z and D become noisy and biased high/low.
- **Too large:** very few blocks are used, estimates have high variance and poor error bars.
- **Just right:** block length >> τ_Z (e.g., 10–50x) while still leaving enough blocks to compute statistics.

If users set it, then it has to be paired with guardrails. A sensible rule of thumb is:

- Choose N_{block} so $N_{block} \cdot \Delta t \geq 10\ \tau_Z$ (preferably 20-50 τ_Z)
- Ensure at least 10–20 total blocks for a stable mean and error

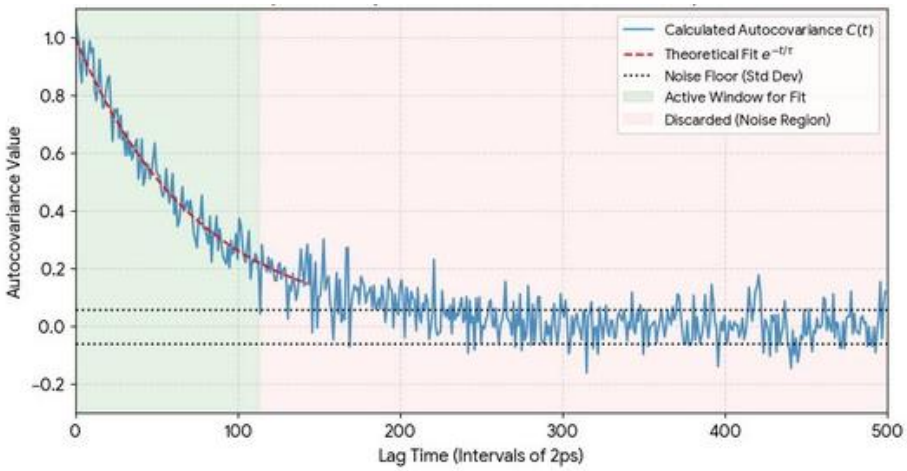


Figure 2. A graphical interpretation of the Hummer method analysis.

NOTE: This alternative approach is statistically superior to a fixed-interval method because it scales the observation window to the actual physical behavior of the molecule. While the Hummer publication uses a fixed number of 500 intervals, this method uses adaptive sampling.

3.8. Acid-Base Equilibria and Thermodynamic Corrections

A defining characteristic of pharmacological compounds is their propensity to ionize. A significant majority of central nervous system drugs exist in an equilibrium between protonated and deprotonated states at a physiological pH of 7.4. The partitioning of a charged molecular species into the low-dielectric, hydrophobic core of a lipid bilayer incurs a massive thermodynamic penalty, manifesting as an insurmountable free energy barrier in the PMF profile. Consequently, passive permeation is frequently dominated by the minor fraction of the uncharged, neutral species, even if that species exists in trace amounts in the bulk aqueous phase.

The software implements a mathematically rigorous correction to account for this dual-state permeation, based on methodologies detailed in contemporary biophysical literature. The documentation clearly outlines the necessary thermodynamic shift, defining the neutralization free energy offset using the Henderson-Hasselbalch equation. The script calculates the exact neutralization energy depending on the specified compound type: For acidic compounds, the shift is calculated as:

$$\Delta G_{neutralize} = (pH - pK_a) \cdot k_B T \cdot \ln(10)$$

For basic compounds, the shift is calculated as:

$$\Delta G_{neutralize} = (pK_a - pH) \cdot k_B T \cdot \ln(10)$$

The software then applies this Δ*G* offset to the PMF of the neutral species, shifting the entire free energy profile relative to the bulk water baseline. The resistivity of the charged pathway is computed

from the original ionized PMF and the charged umbrella sampling diffusion data. Concurrently, the resistivity of the neutral pathway is computed from the shifted neutral PMF and the neutral umbrella sampling diffusion data. Finally, the software combines these two distinct permeation pathways utilizing a parallel resistance model:

$$R_{combined}(z) = \frac{R_{charged}(z) \cdot R_{neutral}(z)}{R_{charged}(z) + R_{neutral}(z)}$$

The total corrected effective permeability is derived by integrating this combined resistivity profile across the membrane. The implementation and theory display perfect parity regarding this complex thermodynamic cycle, demonstrating a robust approach to predicting the permeability of ionizable compounds.

3.9. Error Diagnostics and Data Visualization

The software features critical error diagnostics. Molecular dynamics simulations, particularly umbrella sampling across dense lipid bilayers, are highly prone to sampling failures. If a simulation is too short, or if a specific umbrella window is trapped in a localized energy minimum, the extracted position-dependent diffusion coefficient may approach zero, or mathematical artifacts may result in negative local resistivities. If the integration of the resistivity profile yields a negative or infinite effective permeability (a physically impossible scenario) the software immediately aborts the logBB calculation and triggers a deep diagnostic routine. The codebase iterates through the resultant pandas DataFrame, actively identifying and logging specific trajectory windows where the diffusion coefficient falls below a critical threshold ($D(z) < 1 \cdot 10^{-4}$ nm^2/ps). It identifies windows yielding negative resistivity, and flags potential statistical outliers where the local resistivity exceeds 1000x the median membrane resistivity. Furthermore, the software performs an automated temporal analysis, comparing the total length of the molecular dynamics simulation against the mean correlation time $\tau(z)$. If the simulation length is less than twenty times the mean correlation time, the software outputs a specific recommendation advising the user that the trajectories are too short to yield statistically converged thermodynamic properties. This level of automated, intelligent troubleshooting is executed with remarkable thoroughness in the code.

3.10. Graphical Overlays and Clipboard Integration

The software generates basic plots for the diffusion profile, the PMF and a side-by-side comparison by utilizing matplotlib to render two-dimensional graphs embedded directly into the Tkinter layout via the FigureCanvasTkAgg backend. When a compound is evaluated under the acid-base pKa correction mode, the software does not simply generate static plots. It dynamically generates overlaid graphics, displaying the original neutral PMF alongside a dashed red line representing the pK_a-shifted PMF, visually demonstrating the ΔG neutralization offset. It also generates a separate, color-coded diffusion profile for the neutral species alongside the ionized profile. Additionally, the code features a highly robust, custom right-click context menu bound to the matplotlib canvases. This menu allows users to seamlessly save publication-ready images or copy the rendered graphs directly to the operating system clipboard. The program implements a

sophisticated cross-platform fallback mechanism for this clipboard integration, utilizing win32clipboard native API calls on Windows, osascript on macOS, and attempting both xclip and wl-copy on Linux environments.

In summary:

The key structural and algorithmic achievements in the software architecture include:

- The exact replication of the log-linear autocovariance methodology by Hummer for extracting the relaxation time τ_z and calculating the localized diffusion coefficient $D(z)$, coupled with the proper handling of the requisite noise floor truncation.
- The implementation of an advanced statistical scoring algorithm to dynamically optimize block-averaging intervals, thereby neutralizing the primary source of user error in diffusivity profiling and adapting seamlessly to the heterogeneous viscosity gradients inherent in lipid bilayers.
- The highly accurate rendering of the parallel pathway permeation model, appropriately combining charged and neutral potential of mean force profiles offset by the precise Henderson-Hasselbalch neutralization free energy penalty, $\Delta G_{neutralization}$.
- The integration of comprehensive diagnostic safeguards capable of automatically detecting biophysical anomalies, such as negative membrane resistivity states or non-converged stochastic correlation functions.

Appendix. Practical considerations

a) The PMF_[*drug name*].xvg file should be normalized relative to the bulk solvent prior to its analysis. This requires that the free energy must converge to zero at a distance well beyond the lipid/water interface. In the imipramine example, this is shown in the lower right graphs of Figure 2, where the *ΔG* value starts to oscillate around 0 at a *z*-value of approximately 26Å. This normalization cannot be applied universally, as it depends on each MD system construction and the dimension of the bilayer.

The user must evaluate the free energy changes and decide at which distance from the lipid/water interface the ΔG value converges to or fluctuates around a stable value, which is to be arbitrarily taken as 0 kcal/mol. A Gromacs wham sample command for calculating the PMF is provided:

```
gmx wham \
 -it tpr-files.dat \
 -if pullf-files.dat \
 -o "323_b0_${SYSTEM}.xvg" \
 -hist "323_b0_hist_${SYSTEM}.xvg" \
 -bsres "323_b0_bs_${SYSTEM}.xvg" \
 -bsprof "323_b0_bsprof_${SYSTEM}.xvg" \
 -temp 323 \
 -unit kCal \
 -b 0 \
 -bins 200 \
 -zprof0 x.x \
 -nBootstrap 200
```

Value where *ΔG* converges to zero. Distance units in Gromacs are nanometers (nm).

b) The application utilizes hardcoded subdirectories for locating and opening the necessary umbrella sampling and PMF files. Thus, in Figure 3 an example is shown, demonstrating the proper file hierarchy involved to imipramine theoretical logBB estimation.

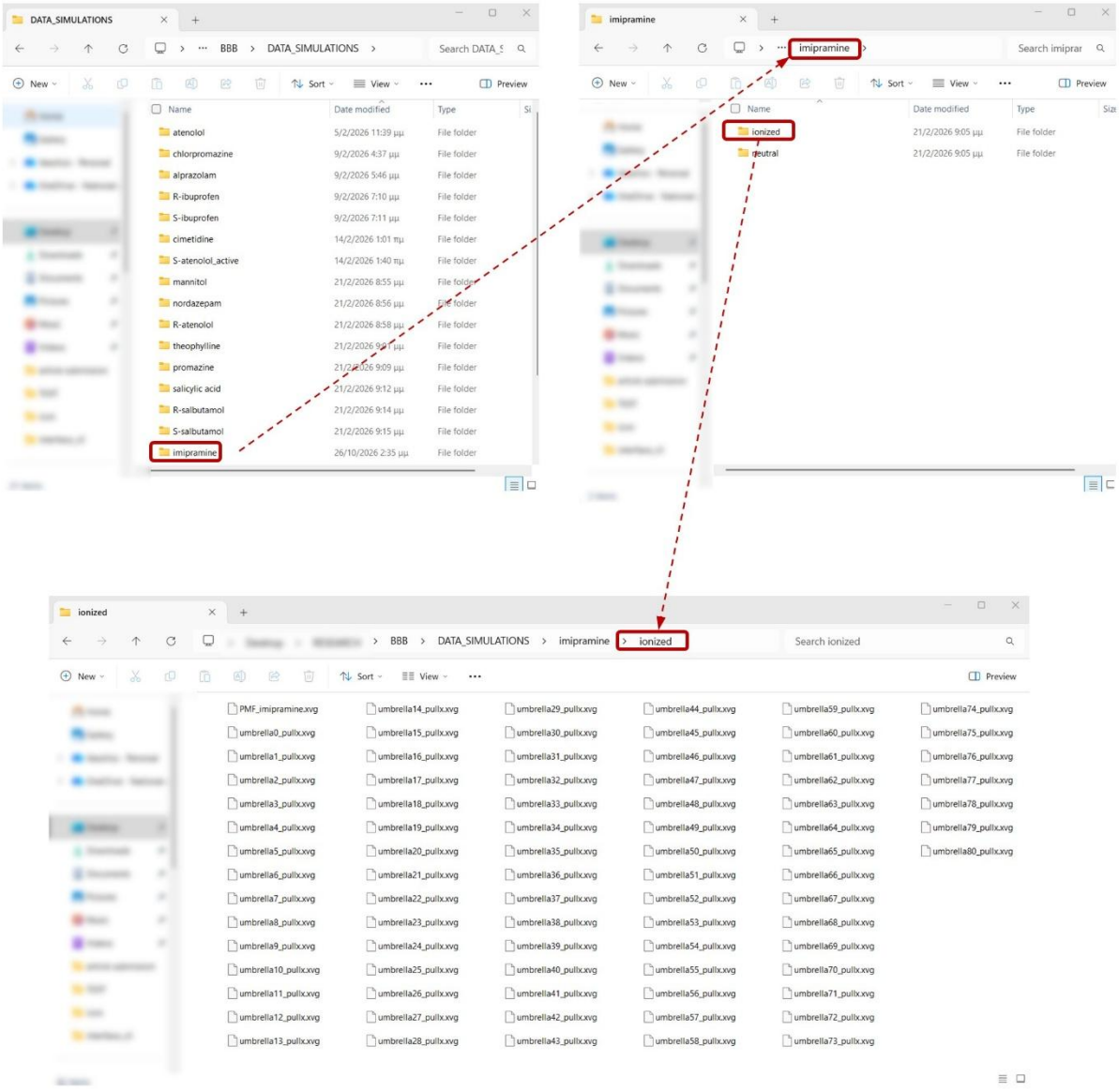


Figure 3. Example of the required directory structure ensuring standard MD3B functionality.

Credits:

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Resources: National and Kapodistrian University of Athens, Department of Pharmacy and National Infrastructures for Research and Technology S.A. (GRNET S.A.) National HPC facility ARIS project ID pa260403

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